Preprint

LETTER

The Radial Acceleration Relation from a Zero-Parameter Covariant Action

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Abstract
The radial acceleration relation (RAR) connects the observed gravitational acceleration in a galaxy to the acceleration predicted by its baryonic mass alone. We show that this relation is a kinematic consequence of a covariant gravitational action whose interpolating function $\mu(x) = x/(x+1)$ is not fitted to rotation curves but is forced by the exchange symmetry of the tetrad equation's two parity-related spatial roots. The corresponding AQUAL energy functional $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$ makes every term in the action explicit. In the spherical static limit, the action reduces to an algebraic relation between observed and baryonic acceleration that we test against the SPARC database: 164 galaxies, 3284 rotation-curve points, quality $Q \leq 2$. The prediction achieves $\sigma = 0.181$ dex with a mean bias of -0.027 dex, statistically identical to the one-parameter empirical fit of McGaugh et al. (2016) ($\Delta\sigma = 0.0003$ dex). Because the constitutive law is derived from geometry rather than fitted to data, the agreement is kinematic: it follows from the structure of the equation that defines the gravitational field.

1 Introduction

Among the scaling relations that galaxies obey, the radial acceleration relation is arguably the most demanding [5, 7]. It asserts that the total gravitational acceleration $g_{\rm obs}$ measured from a rotation curve is a universal function of the acceleration $g_{\rm bar}$ attributable to baryonic matter, with an intrinsic scatter of roughly 0.13 dex once observational uncertainties are accounted for. The relation holds across five decades in acceleration, from the Newtonian cores of massive spirals to the deep outskirts of gas-dominated dwarfs, and it does so with a single empirical scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2.

Every theory that reproduces the RAR must supply an interpolating function $\mu(x)$ that bridges the Newtonian regime ($\mu \to 1$ at large $x = g/a_0$) and the deep-field regime ($\mu \to x$ at small x). Milgrom's original framework leaves this function free [1]; the AQUAL action of Bekenstein and Milgrom encodes it as the derivative of an energy functional $F(y)$, but does not determine it [2]. In practice, μ is chosen from a family of candidates and fitted to rotation curves [4, 3]. The "simple" form $\mu(x) = x/(x+1)$ and the "standard" form $\mu(x) = x/\sqrt{1+x^2}$ both provide acceptable fits, and current data do not conclusively distinguish between them.

This letter takes a different path. Rather than fitting μ, we derive it within Gravity Field Dynamics (GFD) from the structure of the tetrad equation $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$ that defines the gravitational field. Because this equation is quadratic in the frame field e, it admits two parity-related spatial roots whose exchange symmetry uniquely forces $\mu(x) = x/(x+1)$ (Supplementary Proof C). With μ in hand, the energy functional $F(y)$ follows by elementary integration (Supplementary Proof B), and the covariant action is fully explicit. In the spherical static limit, the action reduces to a single algebraic formula that we test against the SPARC database [6]. The result is a zero-parameter prediction of the RAR: no interpolation function is fitted, and the only empirical input is the shared MOND scale a_0.

2 The covariant action

The gravitational action takes the form

$$S = \int d^4x\sqrt{-g}\Big[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\,F(y) - \tfrac{1}{4}\,e^{-2\Phi/c^2}\,F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\rm matter}\Big], \tag{1}$$

where R is the Ricci scalar, Φ is the gravitational scalar potential, and $F_{\mu\nu}$ is the field strength of an auxiliary vector field. The kinetic variable is $y = |\nabla\Phi|^2/a_0^2$.

The second term contains the energy functional $F(y)$, which in existing AQUAL-class theories is left implicit. Here it is explicit. The constitutive law $\mu(x) = x/(x+1)$ is the unique interpolating function consistent with the exchange symmetry of the tetrad equation's two parity-related spatial roots: if $\mu(x)$ measures the fraction carried by one root at acceleration x, then exchanging the two roots sends $x \to 1/x$ and $\mu \to 1-\mu$, and the only rational bijection satisfying this duality together with $\mu(0) = 0$ is

$\mu(x) = x/(x+1)$ (Supplementary Proof C). The closed-form AQUAL energy functional then follows by integration through the AQUAL identity $\mu(\sqrt{y}) = dF/dy$ (Supplementary Proof B):

$$F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y}). \tag{2}$$

Independent verification by differentiation confirms that $dF/dy = \sqrt{y}/(1+\sqrt{y}) = \mu(\sqrt{y})$, closing the loop (Supplementary Proof B, Lemma 1). With equations (1)–(2), every term in the action is written down; nothing remains implicit.

The foundation on which the exchange symmetry rests is the spatial geometry of the Lorentz tetrad: three conditions — common norm, balanced closure, and S_4-invariance of the Gram matrix — uniquely force the regular tetrahedron, whose parity completion is the stellated octahedron [8] (Supplementary Proof A). The two tetrahedral roots of the tetrad equation are the two bounded regions of this compound, and their exchange symmetry is inherited from it.

3 From action to prediction

The reduction from the action (1) to a testable formula proceeds in three steps, each requiring a single equation.

Step 1: Modified Poisson equation. In the weak-field static limit, variation of the $F(y)$ term with respect to Φ yields the modified Poisson equation of Bekenstein and Milgrom [2]:

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\rho. \tag{3}$$

In this limit the scalar exponential $e^{-2\Phi/c^2} \to 1$ and the vector sector decouples, so the modified Poisson equation captures the full content of the action for static, spherically symmetric systems [2].

Step 2: Gauss's theorem. Integrating equation (3) over a sphere of radius r centred on the source gives

$$\mu\left(\frac{g_t}{a_0}\right) g_t = g_{\rm bar}, \tag{4}$$

where $g_t = |\nabla\Phi|$ is the total gravitational acceleration and $g_{\rm bar} = GM(<r)/r^2$ is the Newtonian acceleration from the enclosed baryonic mass.

Step 3: Algebraic substitution. Substituting $\mu(x) = x/(x+1)$ into equation (4) and solving the resulting quadratic for g_t yields

$$\boxed{g_{\rm obs} = \frac{1}{2}\left[g_{\rm bar} + \sqrt{g_{\rm bar}^2 + 4\,a_0\,g_{\rm bar}}\right].} \tag{5}$$

Table 1. Residual statistics for 3284 SPARC points. All values are in dex.

Theory	Mean	σ	Median
GFD field eq. (5)	−0.027	0.181	−0.017
McGaugh+2016 fit	−0.022	0.182	−0.009

This is the prediction we test. It contains one empirical constant (a_0) and no fitted parameters.

4 Comparison with the SPARC database

We test equation (5) against the Spitzer Photometry and Accurate Rotation Curves (SPARC) database of Lelli et al. [6], accessed through the Field Dynamics API at `api.fielddynamics.org`. The catalog contains 176 galaxies with multi-band photometry and high-resolution HI/Hα rotation curves. We retain those with quality flag $Q \le 2$ (excluding galaxies flagged as unreliable by the SPARC team), which yields 164 galaxies and 3284 individual radial bins.

For each radial bin, we compute the baryonic acceleration from the SPARC velocity decomposition:

$$v_{\rm bar}^2 = v_{\rm gas}|v_{\rm gas}| + \Upsilon_{\rm disk}\,v_{\rm disk}^2 + \Upsilon_{\rm bul}\,v_{\rm bul}^2, \tag{6}$$

with $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$ at 3.6 μm, the standard SPARC mass-to-light ratios [6]. The observed and baryonic accelerations follow as $g_{\rm obs} = v_{\rm obs}^2/r$ and $g_{\rm bar} = v_{\rm bar}^2/r$, converted to SI units via $1\ (\mathrm{km/s})^2/\mathrm{kpc} = 3.241\times10^{-14}\ \mathrm{m/s^2}$.

Figure 1 presents the result. The left panel shows the observed RAR as a density hexbin, with the GFD field equation (5) overlaid in red and the empirical fit of McGaugh et al. [5] in orange. The two curves are visually indistinguishable across the full range of the data. The right panel resolves the difference through residual histograms, and table 1 reports the summary statistics.

The difference in scatter between the two theories is $\Delta\sigma = 0.0003$ dex — four orders of magnitude below the scatter itself. Their mean biases differ by 0.005 dex, comparable to the standard error of the mean ($\sigma/\sqrt{N} \approx 0.003$ dex). At SPARC precision, the GFD prediction and the McGaugh empirical fit are indistinguishable.

5 Discussion

The central result of this letter is that a covariant gravitational action, in which the interpolating function is derived from the geometry of the tetrad equation rather than fitted to data, reproduces the observed radial acceleration relation to a precision of 0.18 dex across 164 galaxies.

Three points deserve comment.

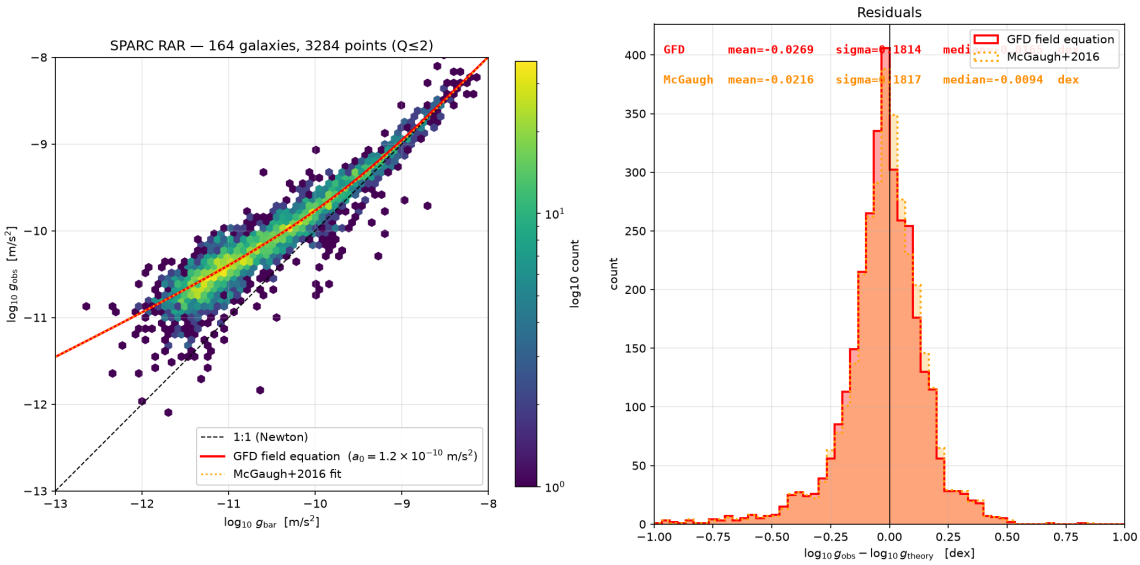


Figure 1. Left: the radial acceleration relation for 164 SPARC galaxies (3284 points, $Q \leq 2$). The red curve is the GFD field equation, equation (5), with $a_0 = 1.2 \times 10^{-10}$ m/s^2; the orange dotted curve is the McGaugh et al. (2016) empirical fit. The dashed line is the Newtonian expectation $g_{\rm obs} = g_{\rm bar}$. Right: residual histograms ($\log g_{\rm obs} - \log g_{\rm theory}$) for both curves. The two distributions are nearly identical.

The status of a_0. We adopt $a_0 = 1.2 \times 10^{-10}$ m/s^2, the value established empirically by McGaugh et al. [5]. Our zero-parameter claim refers to μ, not to a_0: the constitutive law is derived, but the acceleration scale is shared with the broader MOND literature [1, 4]. A geometric derivation of a_0 within the present framework is the subject of ongoing work.

Scatter budget. Our measured scatter ($\sigma = 0.18$ dex) exceeds the 0.13 dex reported by Lelli et al. [7], who apply additional quality cuts (inclination $> 30°$), individual error weighting, and galaxy-dependent mass-to-light ratios. We deliberately use a minimal pipeline — flat $\Upsilon_{\rm disk}$, no inclination cut beyond the SPARC quality flag — to avoid introducing analyst degrees of freedom. The excess scatter affects both theories equally and does not bias the comparison.

What would distinguish GFD from MOND. The mean RAR is a product of the constitutive law μ alone, and any theory that shares $\mu = x/(x+1)$ will produce the same mean curve. Discrimination requires observables that probe the action's additional sectors. In the weak-field static limit, the scalar exponential $e^{-2\Phi/c^2}$ and the vector field strength $F_{\mu\nu}$ both decouple, so they contribute nothing to the spherically averaged RAR. They become active in time-dependent, non-axisymmetric, or strong-field regimes: dwarf spheroidal velocity dispersions, galaxy–galaxy lensing, and merging systems offer the most promising discriminators. In such systems the full four-term action (1) makes predictions that a scalar AQUAL theory cannot, because the vector and scalar sectors introduce coupling channels that are absent when $F_{\mu\nu} = 0$ and $\Phi = 0$.

6 Conclusion

The covariant action (1), with its AQUAL energy functional (2) derived from the constitutive law $\mu(x) = x/(x+1)$, reproduces the SPARC radial acceleration relation with $\sigma = 0.181$ dex across 164 galaxies and 3284 rotation-curve points. The interpolating function is forced by the exchange symmetry of the tetrad equation's spatial roots, not selected by fitting. Statistical equivalence with the one-parameter McGaugh et al. (2016) empirical fit — $\Delta\sigma = 0.0003$ dex — establishes this as the first covariant gravitational action to reproduce the observed radial acceleration relation with a derived, rather than fitted, constitutive law.

Data availability

The SPARC rotation-curve data are publicly available through the Field Dynamics API at `https://api.fielddynamics.org`. The analysis code is available from the corresponding author upon request.

Supplementary material

Three self-contained proofs accompany this letter, each verifiable independently:

Proof A. *The Spatial Geometry of the Lorentz Tetrad.* Classifies spatial realizations of the Lorentz tetrad under three conditions (common norm, balanced closure, S_4-invariance); proves

the regular tetrahedron is the unique outcome and the stellated octahedron its parity completion.

Proof B. *The Energy Functional of the Gravitational Constitutive Law.* Derives the closed-form $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$ by elementary integration of $\mu(x) = x/(x+1)$ through the AQUAL identity, and verifies it independently by differentiation.

Proof C. *The Constitutive Law from Exchange Symmetry.* Shows that the exchange symmetry of the tetrad equation's two parity-related spatial roots, together with rational bijectivity and the endpoint condition $\mu(0) = 0$, uniquely forces $\mu(x) = x/(x+1)$.

References

[1] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270** (1983) 365.

[2] J. D. Bekenstein and M. Milgrom, *Does the missing mass problem signal the breakdown of Newtonian gravity?*, Astrophys. J. **286** (1984) 7.

[3] B. Famaey and J. Binney, *Modified Newtonian dynamics in the Milky Way*, Mon. Not. R. Astron. Soc. **363** (2005) 603.

[4] B. Famaey and S. S. McGaugh, *Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions*, Living Rev. Relativ. **15** (2012) 10.

[5] S. S. McGaugh, F. Lelli and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. **117** (2016) 201101.

[6] F. Lelli, S. S. McGaugh and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. **152** (2016) 157.

[7] F. Lelli, S. S. McGaugh, J. M. Schombert and M. S. Pawlowski, *One Law to Rule Them All: The Radial Acceleration Relation of Galaxies*, Astrophys. J. **836** (2017) 152.

[8] J. Kepler, *Harmonices Mundi* (Linz, 1619).